

Large-Scale Visual Language Model Boosted by Contrast Domain Adaptation for Intelligent Industrial Visual Monitoring

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Abstract—Industrial visual monitoring (IVM) is crucial in enhancing the reliability and efficiency of manufacturing processes. Recently, large vision-language models (LVLMs) have demonstrated remarkable semantic understanding and natural language interaction capabilities, which provide a novel solution to IVM. However, LVLMs pretrained on common domains lack specific knowledge for IVM scenarios, causing insufficient adaptation to industrial image patterns and specialized textual corpora. In this article, we deeply studied the adaptation of LVLMs to IVM and proposed DefectGLM. First, we proposed the first large-scale multimodal wafer dataset as a reliable data basis for model domain generalization. Second, this model employs low-rank adaptation-based contrast visual adaptation to align with industrial image patterns and utilizes vision-language instruction tuning for professional knowledge alignment. DefectGLM is the first large-model-based wafer image recognition model, and can accurately identify 36 types of wafer defects and provide appropriate text descriptions. DefectGLM provides a new solution for the development of industrial large models.

Index Terms—Defect detection, industrial visual monitoring (IVM), large vision-language model (LVLM), semiconductor manufacturing.

I. INTRODUCTION

IN MODERN industry, the manufacturing process tends to be more sophisticated and complex, which raises higher requirements against automation, reliability, and safety [1], [2], [3], [4]. To this end, industrial visual monitoring (IVM) aims to integrate advanced visual sensors and artificial intelligence technologies to oversee and optimize various facets ranging

from production lines to critical infrastructure. An effective IVM method is capable of identifying potential issues, discovering surface anomalies, and recognizing defection patterns [5], [6], [7]. The introduction of IVM enables predictive maintenance and optimization of resource utilization, which shows an important area of applied study.

In recent years, as an efficient data-driven approach, deep learning methods have attracted the attention of researchers, demonstrating excellent application potential in practical monitoring scenarios. It generally trains a neural network with industrial monitoring data, so that the model can identify the anomalies or defection patterns in an end-to-end manner. Convolutional neural networks (CNNs), one of the popular network architectures in computer vision, can extract visual features using convolution kernels capable of recognizing complex, multilevel patterns. Many advanced variants of CNNs have been proposed recently to tackle more complex tasks, e.g., AlexNet [8], VGG [9], ResNet [10], and DenseNet [11]. Additionally, the appearance of transformer architecture also leads to a breakthrough in visual pattern recognition. Methods like Vision Transformer (ViT) [12] and Swin-Transformer [13] introduced attention mechanisms to extract features that are most relevant to an image while suppressing irrelevant details.

These data-driven approaches have achieved great success in IVM scenarios like anomaly detection and defect recognition. However, they still face some limitations in further practical application. The out-of-distribution problems are the first concern against classic deep learning methods. When facing practical scenarios where the data distribution is diverse and complex, these methods trained on small datasets typically face generalization problems [14]. Second, classic deep learning methods are sensitive and unrobust to image noise, which could affect the final diagnosis performance [15]. Finally, these classic vision methods can only output discrete and probability-oriented results, which lack sufficient human interaction.

More recently, the pretrained large language models (LLMs) like ChatGPT, LLaMA [16], and ChatGLM [17] have demonstrated powerful comprehension and dialogue ability to natural language. These LLMs are pretrained with enormous textual data from the Internet and possess a degree of world knowledge from text modality. Besides the language modality, the emergence of the pretrained large vision-language models (LVLMs) has bridged the modality gap across information between images

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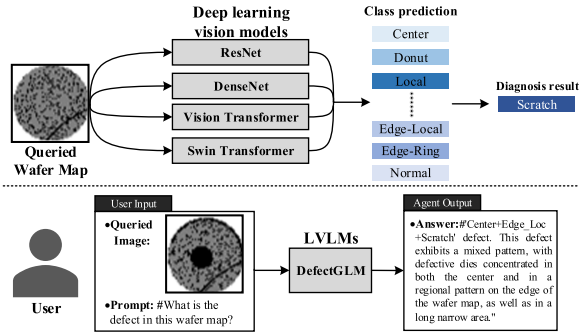


Fig. 1. Difference between existing vision methods and LVLMs in wafer defect recognition scenarios. The vision methods can only output discrete diagnosis results. DefectGLM is an agent that allows for both wafer image and textual instruction input, which can generate recognition results in natural language.

and texts. LVLMs demonstrate a series of advantages, including image semantic understanding, noise robustness, and language interaction capabilities. Therefore, the emergence of LVLMs offered a novel opportunity for creating an IVM agent that can effectively recognize defection patterns and generate diagnosis responses for operators in natural language. The difference between classic deep learning methods and LVLMs is highlighted in Fig. 1. However, LVLMs pretrained on common-knowledge domains are short of proven to be insufficient in obtaining professional domain knowledge for IVM scenarios. Specifically, the absence of domain knowledge includes two aspects. First, LVLMs lack the adaptation of industrial image patterns in specific scenarios. Industrial images have discrepant data distributions and vision patterns compared to common images, which may lead to unsatisfactory image modeling performance in IVM tasks. Second, LVLMs lack alignment with professional knowledge, as the jargon and concepts in IVM scenarios are highly specialized. Consequently, the model may fail to follow instructions and generate the expected diagnostic responses. Therefore, these biases from discrepant vision patterns and different text corpora could cause knowledge absence in LVLMs, resulting in undesirable performance when directly utilized in professional IVM scenarios.

To tackle the knowledge absence problem, we explore adapting LVLMs to professional IVM scenarios and propose DefectGLM, a novel vision-language IVM solution. In this article, we proposed the first large-scale multimodal wafer dataset as a reliable data basis for model domain generalization. Moreover, DefectGLM introduces a simple yet effective two-stage adaptation strategy to compensate for the absence of professional domain knowledge. Specifically, we employ low-rank adaptation (LoRA)-based contrast visual adaptation in the first stage, which enables DefectGLM to adapt specific visual patterns and improve the industrial image modeling capability. Furthermore, a visual language instruction tuning is introduced in the second stage to align DefectGLM to professional domain knowledge, which enhances the instruction-following ability to generate satisfactory diagnosis responses. We utilize the proposed strategy to fine-tune our DefectGLM with the presented dataset, which successfully adapts LVLMs into professional IVM scenarios.

The contributions of this article are summarized as follows.

- 1) This article explores an efficient application of LVLMs to professional IVM tasks, which tackles the problem of lacking specific domain knowledge within LVLMs in two aspects, including discrepant industrial image patterns and different professional domain knowledge.
- 2) This article introduces DefectGLM, a novel two-stage adaptation strategy for LVLMs in IVM. It utilizes LoRA-based contrast visual adaptation and vision-language instruction tuning to enhance defect pattern recognition and professional knowledge alignment. DefectGLM achieves over 99% accuracy in wafer defect recognition, demonstrating state-of-the-art performance.
- 3) The presented multimodal wafer dataset and our DefectGLM will be released for facilitating related IVM research.

The rest of this article is organized as follows. Section II discusses the related work. Section III introduces the proposed DefectGLM method. In Section IV, the experiment results and discussions will be detailed. Finally, Section V concludes this article.

II. RELATED WORKS

A. Deep Learning Based IVM Methods

Recently, many deep learning approaches have been utilized to process and recognize the defect and anomaly pattern in IVM scenarios. CNN is a well-known architecture, which is widely applied in IVM. Wang et al. [18] developed a deformable CNN architecture to extract high-qualified features within the wafer maps for defect pattern recognition. Staar et al. [19] constructed a triplet CNN to inspect industrial surface anomalies. Kim et al. [20] used a CNN-based shot detector to recognize wafer defect objects, which can further localize the defect information. Yu and Liu [21] employed a convolutional autoencoder for wafer defect detection with a novel 2-D principal component convolution kernel. Napoletano et al. [22] introduced a CNN, self-similarity-based method, and scanning electron microscope to detect and localize the anomalies in nanofibrous materials. More recently, other deep learning architectures, e.g., transformers have attracted attention in IVM scenarios. Nafi et al. [23] trained a Swin-Transformer with mixed-type wafer data and achieved great performance in defect recognition. Mishra et al. [24] presented a transformer-based Gaussian mixture density network to capture the anomaly patterns within the industrial images. Hsu et al. [25] combined the ViT and a novel cluster method to analyze wafer map defects. There are still some methods that use U-Net or autoencoder architecture to solve IVM problems. These methods achieved significant success but also faced some problems of generalization, robustness, and interpretability.

B. Large Vision-Language Models

While LLMs have demonstrated powerful performance in many NLP tasks, LVLMs have further extended their abilities into the vision-language modality. Li et al. [26] proposed

BLIP-2, which used a learnable querying transformer to bridge the modality gap between vision and language. Liu et al. [27] introduced LLaVA, which is trained with instruction tuning, which incorporates a learnable projection matrix to convert visual features into the language embedding space. Zhu et al. [28] proposed a MiniGPT4 that utilizes a linear projection to align visual representation with an LLM named Vicuna. Du et al. [17] presented VisualGLM, which utilized the BLIP-2 framework to connect pretrained ViT and ChatGLM, and it was trained with 300M image-text data. These methods demonstrate remarkable capabilities in handling multimodality tasks. The work using vision-language models and contrastive learning with DefectGLM for IVM addresses the gap in adapting LVLMs to specialized industrial scenarios. Ranasingh et al. [29] enhanced vision-language representations by integrating perceptual grouping principles, improving feature alignment and robustness. Gao et al. [30] proposed a pyramid architecture for hierarchical feature alignment, capturing various levels of detail and significantly outperforming existing models in vision-language tasks. Luo et al. [31] introduced contrastive learning and denoising mechanisms to improve the alignment of video frames and textual descriptions, boosting performance on video-language benchmarks. Unlike the general enhancements presented in the earlier works, DefectGLM focuses specifically on IVM, a crucial area for enhancing manufacturing process reliability and efficiency.

However, these models pretrained with common data, may lack specific knowledge in the professional domain, leading to undesirable performance when simply applied to IVM scenarios. To adapt LVLMs to professional tasks, we introduce a novel two-stage adaptation method and propose DefectGLM as a novel vision-language solution for defect recognition.

C. Parameter Efficient Fine-Tuning

To adapt a large-scale model to a specific downstream task, parameter-efficient fine-tuning (PEFT) methods are considered a more efficient solution compared to full fine-tuning. PEFT introduces only a few learnable parameters during fine-tuning, significantly reducing computation costs. Recently, Houlsby et al. [32] proposed adapter tuning, which adds learnable adapter modules to the network architecture to adapt to new tasks. This method reduces fine-tuning costs but increases network depth. Subsequently, Hu et al. [33] introduced LoRA to build a learnable low-rank matrix parallelly in the model, which avoids the increase of network depth. Additionally, Yin et al. [34] employed the tiny adapter with low-rank synthesis to improve the adapter tuning performance. Besides, Dettmers et al. [35] utilized quantization mechanics to optimize the LoRA approach, reducing memory usage without sacrificing performance. Different from the above methods, this study focuses on the generalization and adaptation of multimodal large models to industrial fields and proposes a two-stage adaptation strategy. This strategy uses LoRA-based contrast visual adaptation and visual language instruction tuning methods, which optimize the visual representation module and language generation module to achieve the best understanding of industrial data, and finally comprehensively align the visual

module and language model to achieve an accurate understanding of industrial image data.

III. METHODOLOGY

DefectGLM represents an innovative vision-language solution for addressing IVM tasks by analyzing defect patterns and providing natural language interactions for operators. To tackle the absence of specialized domain knowledge within LVLMs, we propose a two-stage adaptation strategy, comprising contrast visual adaptation and vision-language instruction tuning. The primary objective of contrast visual adaptation is to compensate for the visual pattern gap between Internet common images and professional industrial images. This is achieved by utilizing LoRA adapters and contrastive representation techniques, which enhance the industrial image modeling capability of the DefectGLM. Vision-language instruction tuning aims to align the LVLMs to professional domain knowledge so that the DefectGLM can comprehend specific instructions and therefore generate expected instruction-following responses. In our implementation, we use a pretrained ViT as the visual encoder and a pretrained ChatGLM-6B as the LLM text decoder. We utilize the proposed two-stage adaptation method to fine-tune the pretrained model on a multimodal wafer defect dataset. The generation process of the dataset will be detailed in Section III-D. The DefectGLM can analyze industrial defect patterns and generate instruction-following responses.

A. Network Architecture

The network architecture of the proposed DefectGLM is shown in Fig. 2. Mainly, DefectGLM consists of three parts, a visual encoder for processing vision modality, a querying transformer for bridging the modality gap, and a language decoder for processing and generating language modality. Specifically, the visual encoder uses a pretrained ViT to extract visual representation from the defect patterns. The ViT contains a convolutional projection layer and stacked multihead attention blocks to convert industrial images into patch tokens and then extract multilevel features. The querying transformer is an intermediate module designed to project extracted visual information into language embedding spaces. This lightweight transformer takes the learned tokens as queries and the extracted visual features as keys and values to perform a cross-attention mechanism, which captures text-related visual information from the visual features. Its primary function is to align visual semantics with their corresponding linguistic counterparts, projecting the visual features from a dimensionality of 768 to 409, making them suitable for input into LLMs. For language modality, the language decoder utilizes a novel LLM, i.e., ChatGLM-6B that is pretrained on an enormous text corpus and is capable of understanding and generating natural language. The details of the network architecture are summarized in Table I.

In the fine-tuning process, we freeze all the parameters of the pretrained image encoder and language decoder to reduce training costs and prevent catastrophic forgetting of pretrained knowledge. Furthermore, we plug LoRA into the frozen ViT and

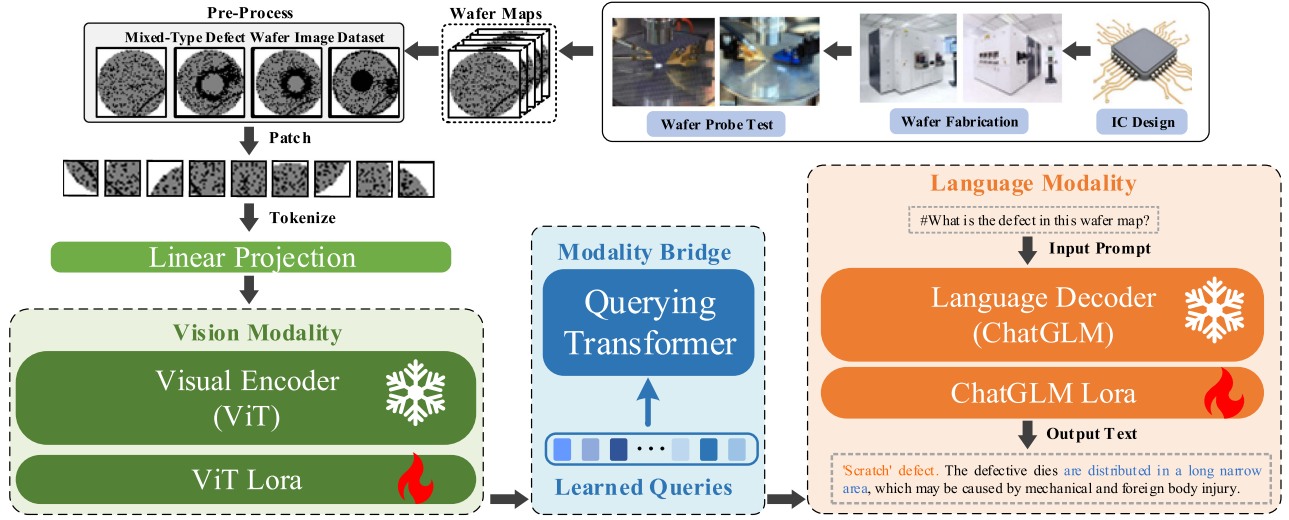


Fig. 2. Network architecture of the DefectGLM. DefectGLM consists of a pretrained ViT as a visual encoder for extracting wafer visual features, a coupled querying transformer for mapping visual features into the language embedding space, and a ChatGLM as an LLM decoder for generating instruction-following responses.

TABLE I
MODEL ARCHITECTURE DETAILS

Model	Depth	Width	MLP	Heads	Params
ViT	39	1408	6144	16	0.96B
Q-Former	12	768	3072	12	0.11B
ChatGLM	28	4096	16384	32	6.20B

ChatGLM as the learnable parameter matrixes. We only update the LoRA parameters when adapting LVLMs to IVM tasks. The utilization of LoRA can help visual adaptation of specific pattern representations for ViT and language instruction understanding for ChatGLM.

B. LoRA-Based Contrast Visual Domain Adaptation

The visual encoder of the proposed DefectGLM is a ViT backbone that is pretrained with huge image data in the common-knowledge domain. However, the direct utilization of freezing ViT in IVM tasks will lead to undesirable performance, which may be caused by discrepant visual patterns in specific industrial scenarios. The pretrained ViT is capable of processing common images but still lacks adaptation of professional visual patterns in industrial scenarios. Therefore, it is essential to perform visual pattern adaptation to the pretrained visual encoder for better extracting industrial image representations to that improve the image modeling capability. The process of contrast visual adaptation is shown in Fig. 3(a).

Here, we utilize the contrastive representation and PEFT strategy LoRA to adapt the pretrained visual encoder to professional IVM tasks. LoRA uses trainable rank decomposition matrixes that can be inserted into any self-attention layers for the pretrained ViT to adapt downstream works. Specifically,

given one weight matrix $W_0 \in \mathbb{R}^{d \times k}$ and input matrix $x \in \mathbb{R}^{b \times d}$, the forward result constrained by LoRA is $W_0x + BAx$, where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ are two low-rank decomposition matrixes, and the rank parameter satisfies $r \ll \min(d, k)$. For the self-attention mechanism in the pretrained ViT, first given three weight matrices, query, key, and value $W_q \in \mathbb{R}^{d_{\text{Hidden}} \times d_k}$, $W_k \in \mathbb{R}^{d_{\text{Hidden}} \times d_k}$, and $W_v \in \mathbb{R}^{d_{\text{Hidden}} \times d_v}$, the attention output is computed as

$$Q = W_q x + B_q A_q x; K = W_k x + B_k A_k x; \\ V = W_v x + B_v A_v x \quad (1)$$

$$\text{Attention}(Q, K, V) = \text{soft max} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2)$$

where Q, K , and V are three matrices modified by the LoRA constraint, d_k is the dimension of the hidden representation. In the visual adaptation process, we freeze all the parameters of pretrained ViT and only train the LoRA adapter that only requires much fewer parameters.

To adapt the pretrained ViT to specific visual patterns in the industrial monitoring scenarios, we employ contrastive representation as a self-supervised visual adaptation approach in DefectGLM. The introduction of contrastive representation can force the pretrained visual encoder to adapt specific visual patterns in a self-supervised manner and extract more transferable representations for the following language modality process. Specifically, suppose we have an industrial image dataset $X = \{x_i, y_i\}_{i=1}^n$ that has a total of n image-label pairs. We perform random augmentation to each image and therefore obtain an augmented dataset $X_{\text{aug}} = \{x_i^{\text{aug}}, y_i\}_{i=1}^n$ as the positive example for the corresponding industrial image x_i . Then, both the datasets are fed into the ViT that is modified with LoRA to obtain the original representation $\{z_i\}_{i=1}^{\text{batchsize}}$ and augmented representation $\{z_j^{\text{aug}}\}_{j=1}^{\text{batchsize}}$ from the final output layer. For each

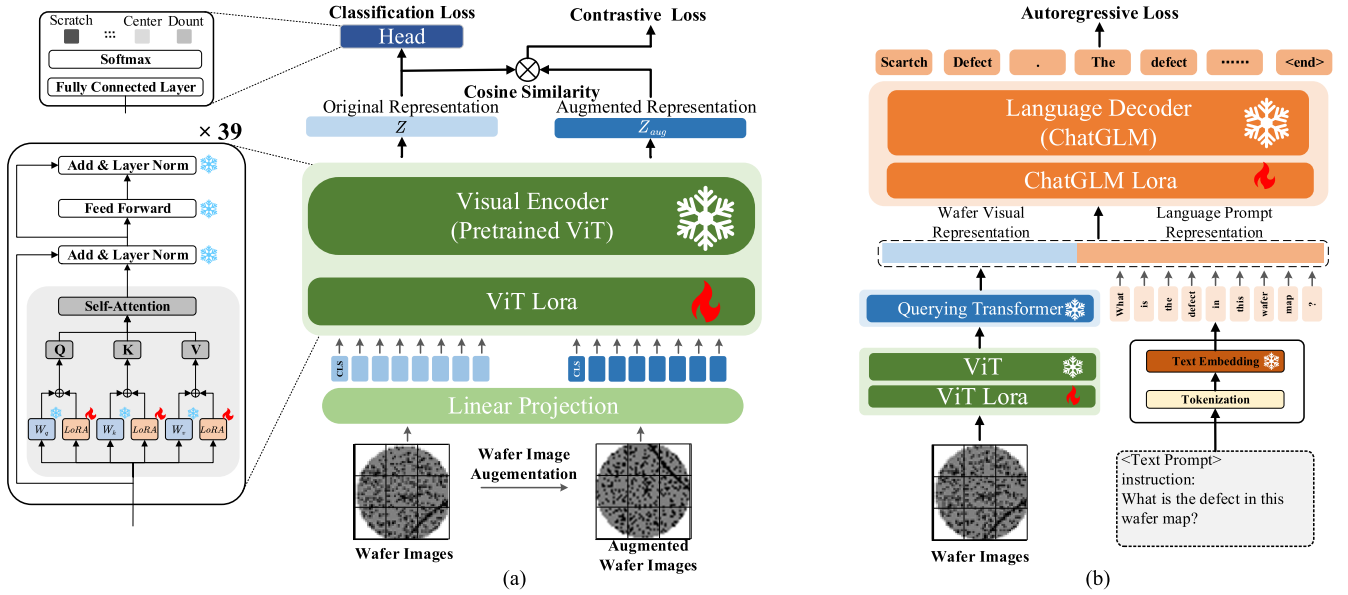


Fig. 3. Two-stage adaptation strategy, including contrast visual adaptation and vision-language instruction tuning. DefectGLM freezes most of the parameters and only updates LoRA parameters during fine-tuning. (a) First stage. (b) Second stage.

representation pair, the InfoNCE contrastive representation loss can be computed as follows:

$$L_r = -\log \frac{\exp(s(z_i, z_j^{\text{aug}})/\tau)}{\sum_j^{\text{batchsize}} \exp(s(z_i, z_j^{\text{aug}})/\tau)} \quad (3)$$

where τ is a temperature threshold (0.1 in this case) and $s(\cdot, \cdot)$ is the cosine similarity score for each pair given as

$$s(z_i, z_j^{\text{aug}}) = \frac{z_i \cdot z_j^{\text{aug}}}{\|z_i\| \|z_j^{\text{aug}}\|}. \quad (4)$$

Additionally, the presented dataset includes ground-truth labels that can be utilized. So, the classification loss is introduced to boost the modeling capability of DefectGLM. Classification loss can assist contrastive loss to learn more category discriminative features, which alleviates the separation of false-negative samples caused by contrastive loss. The classification loss uses a cross-entropy function, which can be derived as

$$L_c = -\sum_{i=1}^{\text{batchsize}} y_i \log(p(x_i)) \quad (5)$$

where $p(x_i) \in \mathbb{R}^{1 \times c}$ is the predicted probability vector and y_i is the one-hot vector. Therefore, the final training goal of the first stage can be the sum of the contrastive adaptation loss and the classification loss.

C. Vision-Language Instruction Tuning

The primary goal of the DefectGLM is to accept industrial visual images and generate concise and reliable natural language responses about diagnosis results. To make the pretrained LLM comprehend the specific task instructions and output desirable language responses based on given industrial images, we employ a vision-language instruction tuning strategy on a generated

image-text dataset to align the domain knowledge for improving the instruction-following capability.

As shown in Fig. 3(b), vision-language instruction tuning leverages a generated image-text dataset as a supervised signal to guide the model to generate expected diagnosis answers based on input industrial images and instruction prompts. Specifically, suppose the industrial visual images $X_v = \{x_i\}_{i=1}^n$ are first fed into the visual encoder ViT $f_v(\cdot)$, which obtains the visual representation $Z_v = f_v(X_v)$. Then, the querying transformer $f_q(\cdot)$ is applied to align the visual representation that is semantic important into the language embedding spaces. Thus, we obtain the output visual tokens $T_v = f_q(Z_v)$. Then, we concatenate the visual tokens with instruction prompt tokens T_q that are obtained by the language embedding layer. We utilize the concatenated multimodal tokens $[T_v; T_q]$ that contain the information of visual images and language instruction as the input of ChatGLM, and therefore predict the text output $y_i = [y_{i,1}, y_{i,2}, \dots, y_{i,T}]$. The autoregressive loss function is utilized here, which can be represented by the negative sum of the predicted probability of $y_{i,t+1}$ given previously predicted text tokens $y_{i,1 \dots t}$. The computation is shown as follows:

$$L = -\sum_{i=1}^{\text{batchsize}} \sum_{t=1}^{T_i} \log[p(y_{i,t+1} | [T_v; T_q], y_{i,1 \dots t}, \Phi)] \quad (6)$$

where T_i is the sequence length of the i th output and Φ is the parameters of the ChatGLM-6B. The autoregressive loss function can make predicted answers close to the ground truth and therefore improve the response performance.

D. Industrial Image and Text Dataset

We utilize the open-source mixed-type wafer dataset published by Wang et al. [18]. The wafer maps are obtained using test probes for each die to test the electrical performance. The

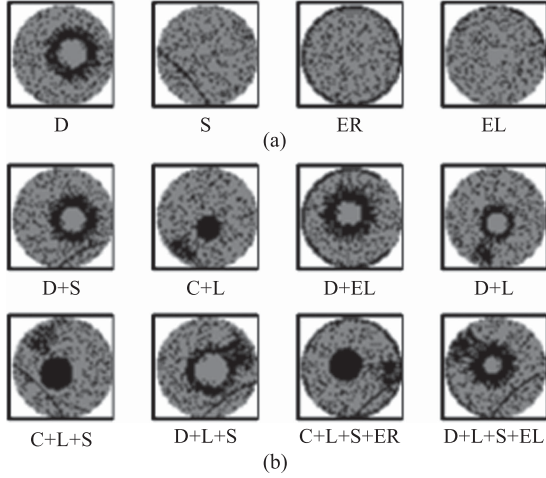


Fig. 4. Mixed-type wafer images with RGB standard. (a) Single defective patterns. (b) Mixed-type defective patterns.

wafer map only contains three different values with 52×52 size, in which 0 denotes a blank plot, 1 presents a normal die, and 2 is a broken die. To adapt the dataset to the input of LVLm, these wafer maps are converted to the standard RGB images with 224×224 size as the input of the visual encoder. The wafer RGB images are shown in Fig. 4. Fig. 4 shows some examples of the single and mixed-type defection patterns. The dataset encompasses 36 distinct defect patterns, including 7 single-type defects, 13 two-mixed type defects, 12 three-mixed type defects, and 4 four-mixed type defects. The single-type defects indicate that only one defect pattern occurs in the wafer map. We considered seven single defect patterns in our study: Normal, Center (C), Donut (D), Edge_Loc (EL), Edge_Ring (ER), Loc (L), and Scratch (S). The “Normal” pattern signifies no defects, whereas C denotes a central circular defect. D is characterized by a circular ring-like defect, EL indicates a local defect region near the wafer’s edge, ER represents a ring defect pattern near the edge, L represents a localized defect region, and S denotes a long strip-like defect.

In addition to these 7 single-type defects, multiple defect patterns may occur in one wafer map, which is a combination of multiple single-type defects. For two-mixed type defects, we considered 13 classes: C + EL, C + ER, C + L, C + S, D + EL, D + ER, D + L, D + S, EL + L, EL + S, ER + L, ER + S, and L + S. These two-mixed type defects feature two single defect patterns in the wafer map. Additionally, we considered three mixed-type defects and four mixed-type defects, including C + EL + L, C + EL + S, C + ER + L, C + ER + S, C + L + S, D + EL + L, D + EL + S, D + ER + L, D + ER + S, D + L + S, EL + L + S, ER + L + S for three mixed-type defects, and C + L + EL + S, C + L + ER + S, D + L + EL + S, and D + L + ER + S for four mixed-type defects.

For text alignment, we first construct a hard prompt as the instruction for all text input, such as *What is the defect in this wafer map?* Then, since the industrial dataset generally lacks text annotations, the description templates are utilized to generate the text label of each wafer image. For instance, “Donut” defect.

The defective dies form a ring structure in this wafer map. The templates consist of two parts: the first gives the diagnosis results, and the second describes the defective patterns illustrated in the wafer images. The generation of text descriptions can also be assisted by ChatGPT or other LLMs.

E. Application Procedure of DefectGLM

The detailed procedure of DefectGLM for fine-tuning LVLms into a professional IVM scenario is shown in this section. The procedure consists of two stages summarized as follows.

Stage I: Contrast visual adaptation

This part includes the following four steps.

- Step 1:* Collect the wafer maps for each defection pattern to generate a fine-tuning dataset.
- Step 2:* Preprocess the collected wafer maps and generate another augmented dataset.
- Step 3:* Both original and augmented wafer images are used as the ViT inputs for representation extraction.
- Step 4:* Use extracted representations to fine-tune LoRA in ViT with contrastive and classification learning.

Stage II: Vision language instruction tuning

This part includes the following five steps.

- Step 1:* Generate a visual instruction dataset with wafer images and text templates.
- Step 2:* Input the wafer images into the pretrained ViT to extract visual representations.
- Step 3:* The extracted visual representations are used as input for the querying transformer to project them into the language embedding space.
- Step 4:* The projected representation is concatenated with the token embeddings of language instruction, and then used as input for ChatGLM.
- Step 5:* The outputs of the ChatGLM and the text labels are utilized to fine-tune the LoRA parameters.

By employing the two-stage adaptation strategy, the DefectGLM can be adapted to professional industrial monitoring tasks.

IV. EXPERIMENTS

A. Experiment Setting

1) Dataset: The experiments were conducted on a diverse wafer dataset comprising 36 distinct categories of defective patterns [18]. To maintain data balance, we allocated 800 wafer images from each category for the training dataset, while reserving 100 wafer images for validation and 50 for testing purposes. In total, our dataset is comprised of 34 200 wafer images, which were utilized for both training and assessing the performance of each method.

2) Evaluation Metric: To evaluate the performance of different models, several evaluation metrics are utilized. Since wafer defection recognition is a classification task, we follow the classic setting of multiclass classification, including mean accuracy, precision, and F1-score to evaluate the defect recognition performance of the model.

TABLE II
DEFECT RECOGNITION RESULTS OF LoRA SENSITIVITY STUDIES

LoRA Rank	Params	Params reduction (%)	Accuracy	Precision	F1-score
8	0.86M	99.988	98.67	98.70	98.67
16	1.59M	99.978	99.11	99.13	99.11
32	2.76M	99.962	98.33	98.41	98.33
64	5.12M	99.929	98.83	98.85	98.83
128	12.68M	99.824	98.78	98.80	98.77

3) Implementation Details: We use the pretrained ViT as the visual encoder of the backbone and utilize the pretrained ChatGLM-6B as the text decoder. We plug the ViT LoRA (VL) at 0, 13, 26, and 38 layers and ChatGLM LoRA (CL) at 0, 19, and 27 layers as the trainable parameters for fine-tuning. The fine-tuning process was conducted on two NVIDIA A40 GPUs throughout 4500 iterations for each stage, resulting in a total of 9000 iterations. The learning rate was set at 1e-3 and a batch size of 32 was employed. To enhance the stability of the training process, we utilized a warm-up phase of 0.2 and implemented a cosine learning rate decay strategy. It is important to note that all pretrained models remained frozen during the training process, and only the LoRA parameters will be updated. This strategy was employed to ensure effective knowledge transfer while fine-tuning the model using wafer images and instructional data.

B. LoRA Sensitivity Results

This section aims to investigate the affection of the rank number of the LoRA adapter in industrial monitoring scenarios. The LoRA rank will directly affect the number of trainable parameters in the DefectGLM, resulting in different adaptation performances on specific scenarios. A larger LoRA rank can lead to more trainable parameters to enable more volume for learning new knowledge but it also requires more training samples and times for convergence. Therefore, investigating the optimal value of LoRA rank is essential for applying LVLMs to specific scenarios. The number of LoRA ranks in the ViT and ChatGLM varies from 8 to 128 to adapt the wafer defect recognition task. We record the number of trainable parameters and the percentage of parameter reduction compared to the full-fine-tuning. The parameter changes and final recognition results under different LoRA rank settings are drawn in Table II.

According to Table II, it is obvious that using LoRA can significantly reduce the total number of trainable parameters by up to 99.98%. The results show that LoRA is efficient in saving memory and computational resources during the fine-tuning process.

Table II shows that the DefectGLM performs best when the LoRA rank is 16 and the corresponding trainable parameters are 1.59 M. We also observe that the performance will first increase and then decrease as increasing the value of the LoRA rank. This could be caused by the small size of the wafer defection dataset, which cannot provide enough training samples for adapting

TABLE III
ABLATION RESULTS OF DIFFERENT COMPONENTS OF EACH COMPONENT

CL	VL	L_c	L_r	Accuracy	Precision	F1-score
✓				7.67	5.59	4.32
	✓			95.72	96.11	95.72
✓	✓			97.44	97.58	97.44
	✓		✓	98.89	98.91	98.89
	✓	✓		98.33	98.40	98.34
✓	✓	✓		98.89	98.91	98.89
✓	✓		✓	98.89	98.91	98.89
✓	✓	✓	✓	99.11	99.13	99.11

more LoRA parameters. Thus, we use a LoRA rank of 16 in the ViT and ChatGLM as the final solution of the proposed DefectGLM.

C. Ablation Results

In this section, we conduct ablation studies to verify the effectiveness of each component in the proposed DefectGLM. We investigate the influence of adjusting vision parameters and the effectiveness of the proposed visual representation adaptation. We only retain CL to perform language instruction tuning and eliminate VL to verify if the LVLMs can obtain desirable performance using pure language techniques without vision adjustments. Then, we continue to explore how the contrast visual adaptation learning loss influences the recognition result. All the ablation studies are conducted with the LoRA rank of 16 for a fair comparison. Notably, L_c and L_r are specifically designed to improve the vision encoder, so we only conduct ablation experiments for L_c or L_r when VL is present.

Table III presents the recognition performance results obtained from the ablation studies. These ablation results highlight several pivotal factors for adapting LVLMs to the complex domain of professional industrial monitoring scenarios. First, the comparison between the first and third rows in Table III, shows that only adjusting language parameters is insufficient for adapting LVLMs. The possible reason could be the absence of visual knowledge within the visual encoder, so it is essential to conduct visual pattern adaptation to adapt specific industrial image patterns. By comparing the second row and the first row, it is further verified the importance of adjusting the vision encoder to adapt industrial image patterns. We found that the accuracy of the third row is 97.44%, which demonstrates that using both CL and VL can help to improve model performance. Then, we investigate how contrast-representation learning and supervised learning contribute to adapting the vision encoder. By comparing the second row and the fourth row, it is obvious that contrast learning provides an efficient solution for the vision encoder to learn defect pattern-related representation. The comparison between the second row and the fifth row shows that supervised learning can also enforce the vision encoder to learn category-discriminative representations. Both these learned representations can contribute to improving the

TABLE IV
DEFECT RECOGNITION RESULTS OF COMPARISON STUDIES

Methods	#Params	Inference Time (ms)	2-Mixed type defect		3-Mixed type defect		4-Mixed type defect		Total	
			Accuracy	F1-score	Accuracy	F1-score	Accuracy	F1-score	Accuracy	F1-score
VGG19	139.7 M	1.7232	98.31	99.06	96.50	97.95	98.31	97.98	97.72	97.72
ResNet18	111.9 M	2.5434	98.46	99.07	97.17	98.38	98.46	97.96	98.00	97.99
ResNet50	235.8 M	5.6259	98.46	99.14	95.33	97.16	98.46	97.72	97.44	97.43
DenseNet	6.9 M	13.3998	98.92	99.45	98.17	98.82	98.92	98.23	98.61	98.61
ViT	85.8 M	4.1768	98.31	99.06	97.17	98.46	98.31	98.74	98.17	98.16
Swin-Trans	86.7 M	20.3840	97.69	98.67	95.67	97.34	97.69	97.20	97.06	97.05
Swin-Trans V2	86.9 M	29.7161	97.23	98.35	95.67	97.61	97.23	98.22	97.17	97.16
ConvNeXt	87.8 M	10.0676	98.92	99.45	97.67	98.40	98.92	98.99	98.56	98.55
DefectGLM (Ours)	1.59 M Total 7.2B	1569.5708	99.54	99.77	98.67	98.99	99.54	99.25	99.11	99.11

answer quality of LLMs. Notably, the accuracy results of the fourth row and the seventh row are the same, which the possible reason is that the model may met a performance bottleneck when approaching an accuracy of 98.9%. Finally, the final row achieved the best performance, which shows the classification task can further boost the contrastive training. The possible reason could be that using classification loss can alleviate the separation of negative samples caused by contrast learning. This result shows that contrast representation loss and classification loss are complementary learning approaches that can learn more comprehensive representations of LLMs.

D. Comparison Results

In this section, we conduct comparative studies utilizing the mixed-type wafer dataset in conjunction with other well-established vision methods, including VGG19 [7], ResNet18 and ResNet50 [10], DenseNet [11], ViT [12], Swin-Transformer [13], Swin-Transformer V2 [36], and ConvNeXt [37]. To adapt these methods to the task at hand, we replace their final layers with initialized linear classifiers to facilitate the classification of defect patterns. Subsequently, we select the model with the highest validation performance and assess its defect recognition capabilities on the test dataset.

The results of wafer recognition on the testing dataset are presented in Table IV. Our proposed method consistently outperforms other approaches for all mixed wafer defects, achieving a remarkable overall accuracy exceeding 99.0%. Notably, the convolution-based methods (such as VGG, ResNet, DenseNet, and ConvNeXt) generally exhibit superior performance compared to the transformer-based methods (such as ViT and Swin-Transformer). This disparity can be attributed to the fact that transformer models typically require a more extensive amount of training data to converge and achieve satisfactory performance. This requirement is often challenging to meet in scenarios involving small wafer datasets. This finding underscores the robust generalization capabilities of DefectGLM, which prove effective even in cases with limited dataset sizes.

Additionally, Table IV also compared the trainable parameters and inference time of the different methods. We use one NVIDIA A40 (48 GB) GPU with an Intel Xeon Platinum 8358P CPU to evaluate the average inference time for each image in the test dataset. Our proposed DefectGLM stands out as a considerably

larger model than the classic vision models, boasting a total parameter count of 7.2 billion. However, it is essential to note that only 1.59 M parameters require updating during the training process, a significantly smaller number than other models.

The results given in Table IV show that the convolutional and ViT methods achieve the quickest inference times, which are beneficial because of their small parameter sizes and high parallel architecture. Although DenseNet has fewer parameters, its inference time is significantly longer, because it has larger feature maps with more channels that increase computational demands during convolutional operations. Additionally, the Swin-Transformer's shifting-window attention mechanism, designed to capture hierarchical information, adds to its computational complexity and increases the inference time. DefectGLM exhibits the slowest inference time, which is caused by extensively higher parameter size and recursive text generation mechanism. As a vision-and-language model, DefectGLM generates the text by recursively predicting the next token, which the recursive mechanism naturally slows down the inference speed.

To mitigate the issue of slow inference time and enhance efficiency in real-world applications, several strategies can be employed. First, applying quantization techniques to convert the model to INT8 or even INT4 data types can conserve computational resources. Second, utilizing batch inference can speed up the process. Third, model pruning can be used to eliminate redundant parameters, simplifying the model and thereby reducing computational complexity and improving inference speed.

E. Qualitative Results

The examples shown in Fig. 5 showcase the practical application of DefectGLM in wafer defect recognition. We can see that DefectGLM demonstrates high generalization to different text prompts and can generate expected answers. In contrast to previous methods that are limited to providing classification results and lack substantial user interaction, DefectGLM offers a comprehensive language-vision solution for IVM scenarios. Notably, wafer defect recognition is only one case of the DefectGLM; the proposed method can also be utilized in other scenarios, such as anomaly detection. The advantages of DefectGLM can be summarized as follows. First, DefectGLM leverages the powerful instruction-following and image semantic understanding capabilities of pretrained LLMs, thus

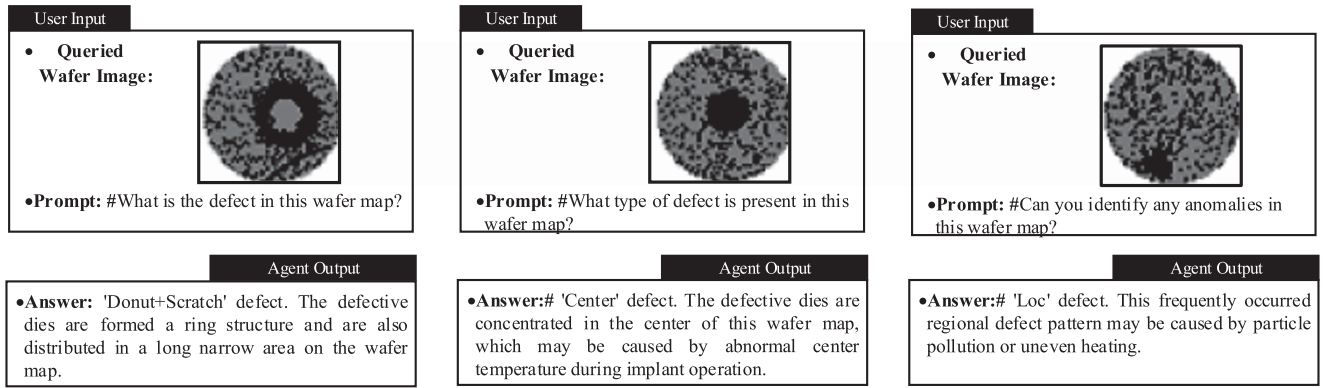


Fig. 5. Qualitative examples of the DefectGLM.

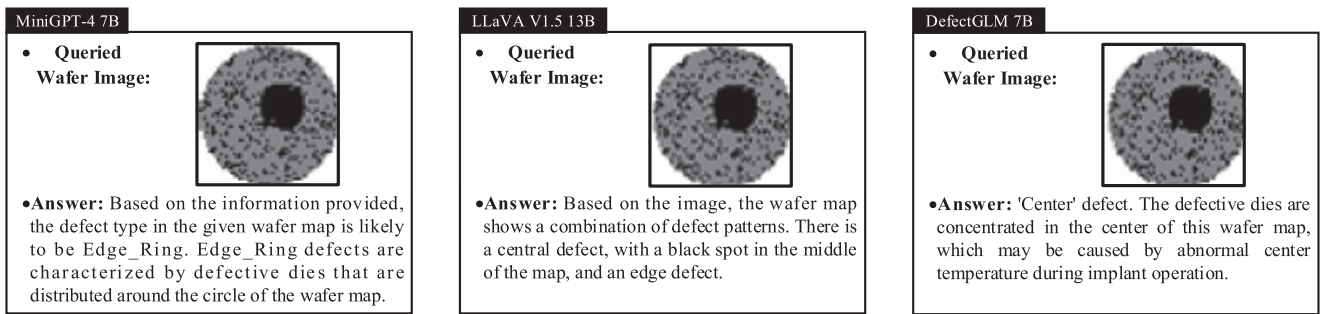


Fig. 6. Comparison results with other LVLM-powered VQA models.

achieving excellent defect pattern recognition performance in IVM scenarios. Second, the DefectGLM can follow prompt instructions and generate concise and clear instruction-following responses, which allows more human interaction with operators. Finally, the DefectGLM represents a novel technique solution path that integrates LVLMs into intelligent IVM tasks.

Additionally, to compare our DefectGLM with other LVLM-powered vision question-answering models, we conducted an experiment to evaluate their answer quality against a wafer map with a central defect. We select MiniGPT-4 7B [28] and LLaVA v1.5 13B [27] as the comparison methods and evaluate the answer from diagnosis correctness and pattern understanding, which the results are shown in Fig. 6. From Fig. 6, MiniGPT-4 shows some ability to recognize the circular defect patterns but outputs an incorrect diagnosis. LLaVA's answer identifies both center and edge defects, correctly recognizing the central black spot in the wafer map, but incorrectly identifying an edge defect that does not exist. Finally, our proposed DefectGLM provides the correct diagnosis and pattern description, demonstrating the effectiveness of the proposed two-stage fine-tuning strategy.

V. CONCLUSION

In this article, we undertook a comprehensive exploration to efficiently adapt pretrained LVLMs to meet the demands of professional IVM scenarios. We introduced an effective two-stage adaptation strategy to address the challenge of domain

knowledge absence in LVLMs. The strategy compensated for the knowledge absence in two aspects. First, the LoRA-based contrast visual adaptation allowed the model to adapt specific industrial visual patterns, thus reducing the knowledge gap between common images and industrial images. Second, vision-language instruction tuning aligned the model to a specific knowledge domain and enhanced its instruction-following ability.

By utilizing the proposed adaptation strategy to pretrained LVLMs on a multimodal wafer defect dataset, a novel vision-language method, DefectGLM, was proposed. The proposed DefectGLM was specialized for IVM scenarios, which could recognize defect patterns and generate instruction-following responses. After evaluation, the DefectGLM showcased remarkable wafer defect recognition performance, which obtained an accuracy of over 99.0%. In our future work, we will widely validate the performance of the proposed method on other general multimodal model structures to extend its applications and values. Finally, to facilitate the IVM research, we will release our DefectGLM and the dataset.

REFERENCES

- [1] S. Yin, S. X. Ding, X. Xie, and H. Luo, "A review on basic data-driven approaches for industrial process monitoring," *IEEE Trans. Ind. Electron.*, vol. 61, no. 11, pp. 6418–6428, Nov. 2014.
- [2] S. Yin, H. Gao, and O. Kaynak, "Data-driven control and process monitoring for industrial applications—Part II," *IEEE Trans. Ind. Electron.*, vol. 62, no. 1, pp. 583–586, Jan. 2015.

- [3] X. Kong and Z. Ge, "Deep learning of latent variable models for industrial process monitoring," *IEEE Trans. Ind. Inform.*, vol. 18, no. 10, pp. 6778–6788, Oct. 2022.
- [4] W. Lu and X. Yan, "Deep double supervised embedding neural network enhancing class separation for visual high-dimensional industrial process monitoring," *IEEE Trans. Ind. Inform.*, vol. 17, no. 9, pp. 6357–6367, Sep. 2021.
- [5] L. Qi, Y. Yang, X. Zhou, W. Rafique, and J. Ma, "Fast anomaly identification based on multiaspect data streams for intelligent intrusion detection toward secure Industry 4.0," *IEEE Trans. Ind. Inform.*, vol. 18, no. 9, pp. 6503–6511, Sep. 2022.
- [6] D. Tabernik, S. Šela, J. Skvarč, and D. Škočaj, "Segmentation-based deep-learning approach for surface-defect detection," *J. Intell. Manuf.*, vol. 31, no. 3, pp. 759–776, Mar. 2020.
- [7] J. Yu and J. Liu, "Multiple granularities generative adversarial network for recognition of wafer map defects," *IEEE Trans. Ind. Inform.*, vol. 18, no. 3, pp. 1674–1683, Mar. 2022.
- [8] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 87–90, Jun. 2017, doi: [10.1145/3065386](https://doi.org/10.1145/3065386).
- [9] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2023, *arXiv:1409.1556*.
- [10] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [11] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 2261–2269.
- [12] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2021, *arXiv:2010.11929*.
- [13] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 9992–10002.
- [14] G. Pang, L. Cao, and C. Aggarwal, "Deep learning for anomaly detection: Challenges, methods, and opportunities," in *Proc. 14th ACM Int. Conf. Web Search Data Mining*, 2021, pp. 1127–1130.
- [15] N. Akhtar and A. Mian, "Threat of adversarial attacks on deep learning in computer vision: A survey," *IEEE Access*, vol. 6, pp. 14410–14430, 2018.
- [16] H. Touvron et al., "LLaMA: Open and efficient foundation language models," 2023, *arXiv:2302.13971*.
- [17] Z. Du et al., "GLM: General language model pretraining with autoregressive blank infilling," in *Proc. 60th Annu. Meeting Assoc. Comput. Linguistics*, 2022, pp. 320–335.
- [18] J. Wang, C. Xu, Z. Yang, J. Zhang, and X. Li, "Deformable convolutional networks for efficient mixed-type wafer defect pattern recognition," *IEEE Trans. Semicond. Manuf.*, vol. 33, no. 4, pp. 587–596, Nov. 2020.
- [19] B. Staar, M. Lütjen, and M. Freitag, "Anomaly detection with convolutional neural networks for industrial surface inspection," *Procedia CIRP*, vol. 79, pp. 484–489, Jan. 2019.
- [20] T. S. Kim, J. W. Lee, W. K. Lee, and S. Y. Sohn, "Novel method for detection of mixed-type defect patterns in wafer maps based on a single shot detector algorithm," *J. Intell. Manuf.*, vol. 33, no. 6, pp. 1715–1724, Aug. 2022.
- [21] J. Yu and J. Liu, "Two-dimensional principal component analysis-based convolutional autoencoder for wafer map defect detection," *IEEE Trans. Ind. Electron.*, vol. 68, no. 9, pp. 8789–8797, Sep. 2021.
- [22] P. Napolitano, F. Piccoli, and R. Schettini, "Anomaly detection in nanofibrous materials by CNN-based self-similarity," *Sensors*, vol. 18, no. 1, Jan. 2018, Art. no. 209.
- [23] T. Nafi, E. Haque, F. Farhan, and A. Rahman, "High accuracy swin transformers for image-based wafer map defect detection," *Int. J. Eng. Manuf.*, vol. 12, pp. 10–21, Oct. 2022.
- [24] P. Mishra, R. Verk, D. Fornasier, C. Piciarelli, and G. L. Foresti, "VT-ADL: A vision transformer network for image anomaly detection and localization," in *Proc. IEEE 30th Int. Symp. Ind. Electron.*, 2021, pp. 1–6.
- [25] Y.-M. Hsu, X. Jia, W. Li, and J. Lee, "A novel quality clustering methodology on fab-wide wafer map images in semiconductor manufacturing," in *Proc. ASME 17th Int. Manuf. Sci. Eng. Conf.*, 2022, vol. 85819, Paper V002T06A022.
- [26] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, 2023.
- [27] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 34892–34916.
- [28] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," 2023, *arXiv:2304.10592*.
- [29] K. Ranasinghe, B. McKinzie, S. Ravi, Y. Yang, A. Toshev, and J. Shlens, "Perceptual grouping in contrastive vision-language models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 5571–5584.
- [30] Y. Gao et al., "PyramidCLIP: Hierarchical feature alignment for vision-language model pretraining," *Adv. Neural Inf. Process. Syst.*, vol. 35, pp. 35959–35970, Dec. 2022.
- [31] J. Luo, Y. Li, Y. Pan, T. Yao, H. Chao, and T. Mei, "CoCo-BERT: Improving video-language pre-training with contrastive cross-modal matching and denoising," in *Proc. 29th ACM Int. Conf. Multimedia*, 2021, pp. 5600–5608.
- [32] N. Houlsby et al., "Parameter-efficient transfer learning for NLP," in *Proc. 36th Int. Conf. Mach. Learn.*, 2019, pp. 2790–2799.
- [33] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," 2021, *arXiv:2106.09685*.
- [34] D. Yin, Y. Yang, Z. Wang, H. Yu, K. Wei, and X. Sun, "1% vs 100%: Parameter-efficient low rank adapter for dense predictions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 20116–20126.
- [35] T. Detmmer, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLORA: Efficient finetuning of quantized LLMs," *Adv. Neural Inf. Process. Syst.*, vol. 36, pp. 10088–10115, 2023.
- [36] Z. Liu et al., "Swin transformer V2: Scaling up capacity and resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 12009–12019.
- [37] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11976–11986.